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AM-740

High Capacity Memory Board Installation Instructions

PDI-00740-00,Rev.A02

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REVISIONS INCORPORATED	
REVISION	DATE
00	June 1987
01	Nov. 1987
02	Dec. 1992

AM-740 High Capacity Memory Board Installation Instructions
To re-order this document, request part number PDI-00740-00.

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1.0 INTRODUCTION

This document gives hardware and software instructions for installing the AM-740 High Capacity Memory Board into an Alpha Micro VMEbus-based computer. We discuss the following topics:

- General product information.
- Step-by-step hardware installation instructions.
- Using the PARITY command.
- Initial testing procedure.

This document contains information on procedures that should be performed by an experienced service technician. If you are not comfortable performing these procedures yourself, please contact an authorized Alpha Micro service technician for help.

2.0 GENERAL PRODUCT DESCRIPTION

The AM-740 contains high capacity dynamic MOS READ/WRITE memory with byte parity that can be installed in any Alpha Micro VMEbus-based computer. It is available in 4MB, 8MB, and 12MB versions. The order number for each version appears in the table below:

Part Number	Description
PDB-00740-00	12MB Memory Board
PDB-00740-01	8MB Memory Board
PDB-00740-02	4MB Memory Board

2.1 Specifications

The basic features of the AM-740 memory board are:

- Bus Interface - Standard VMEbus, 8, 16 or 32-bit transfers
- 4MB, 8MB, or 12MB dynamic MOS read/write.
  (Uses 1 mega x 1 dynamic memory chip.)
- Addressing: Full 24-bit standard addressing (16MB),
  and up to 30 bits (1 giga bytes) of extended addressing.

- Base board address is user selectable in 4MB memory address increments (see Figure 3.)
- Byte parity error detection capability.
- Unique high speed memory clearing function for fast system start-up operation.
- Block transfer capabilities.

2.2 Environmental Specifications

Operating Temperature: 60 F to 130 F (16 C to 55 C)
Humidity: 10 to 90% (non-condensing)

2.3 Power Specifications

One 12MB AM-740 board operates on +5 volts DC with a maximum current draw of 3.5 amps.

2.4 Compatibility/Requirements

The AM-740 board is compatible with Alpha Micro VMEbus-based computers running AM-175, AM-180, AM-185, and AM-185(-50) CPU boards.

If your computer is an AM-1500 series (using an AM-175 CPU), the boot PROM on the CPU board must be revision F00 or later and the monitor level must be AMOS/L 1.3C (157)-2 or later. If your computer is an AM-2000 series (using an AM-180 CPU board) the boot PROM on the CPU board must be revision C00 or later and the monitor level must be AMOS/32 1.0A (154)-2 or later. All versions of AM-3000 series computers (using AM-185 or AM-185(-50) CPU boards) have boot PROMS and operating systems that are compatible with AM-740 memory boards.

Please contact your dealer or an Alpha Micro service company for information if you are unable to resolve a compatibility problem.

3.0 INSTALLATION OVERVIEW

This section contains a brief overview of the hardware installation procedures you will follow to install the AM-740 board into your computer system.

Overview:

1. Selecting address/control jumper configurations.
2. Opening the computer.
3. Installing the AM-740 board.
4. Adding the PARITY statement to system initialization command file.

4.0 AM-740 INSTALLATION

Before installing the AM-740, turn off the power to the computer, but DO NOT unplug the power cable from the system. The power cable keeps the computer grounded.

Prior to handling the AM-740 board(s), or any of the VMEbus-based system boards, GROUND YOURSELF. Static electricity can damage internal components. The best ground is the cabinet that houses the VME boards, so touch the cabinet prior to picking up, installing, or removing them. Do not unplug the power cord, but insure that the power switch is turned OFF. If you unplug the computer cabinet, you will have isolated it from the common earth ground, leaving it vulnerable to static discharge.

4.1 Preparing the AM-740 Board For Installation

Refer to Figure 1, the AM-740 component layout drawing, for the location of the user configurable jumpers described in this section.

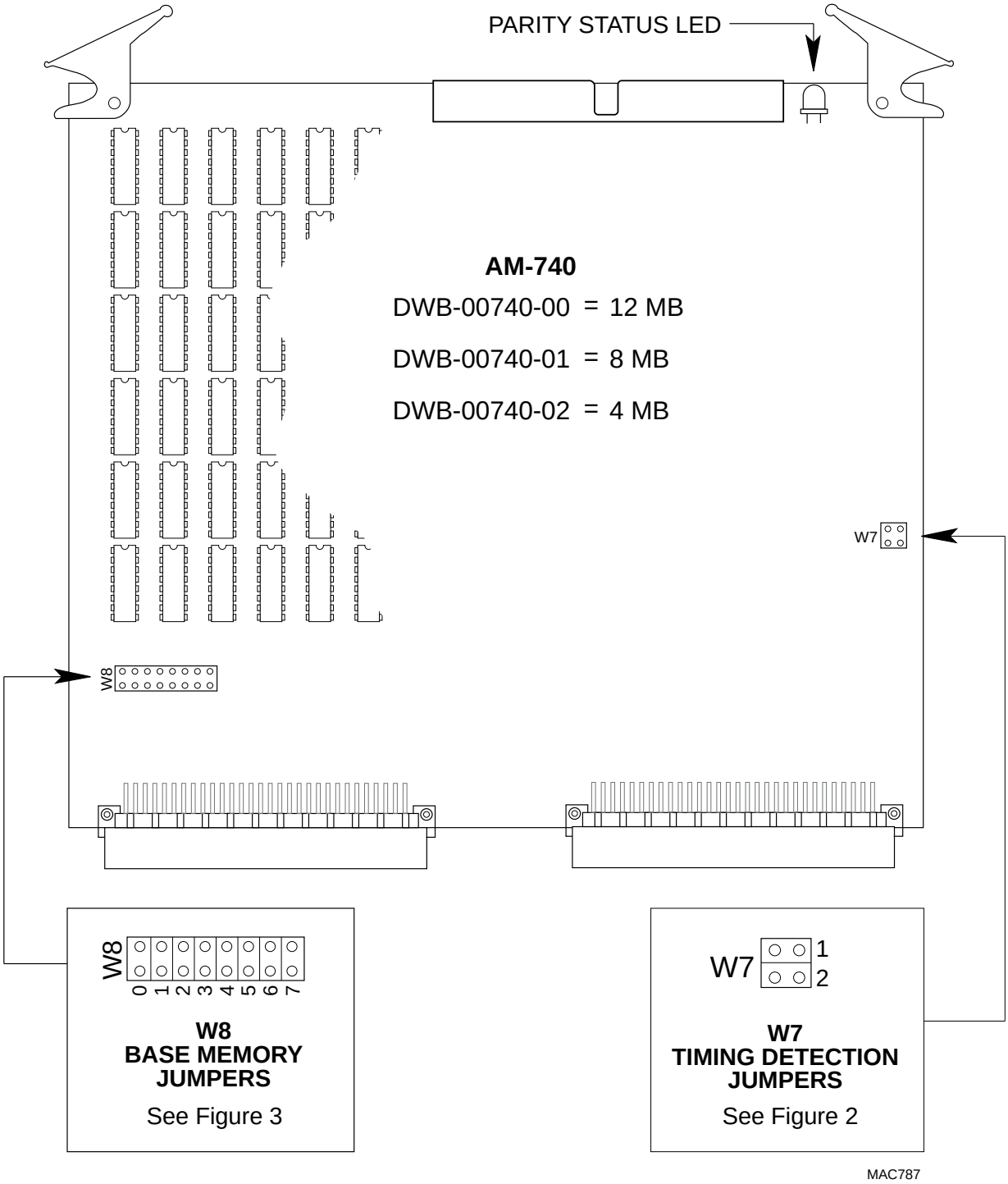
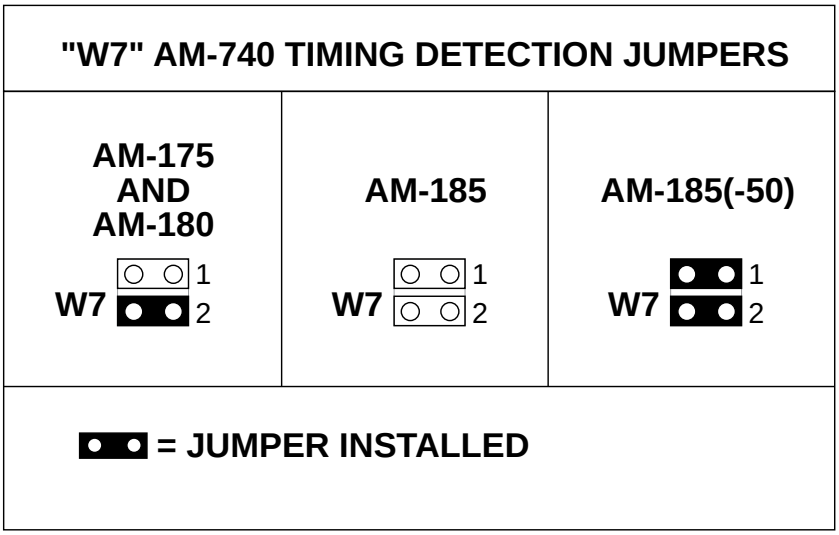


Figure 1. AM-740 Component Layout

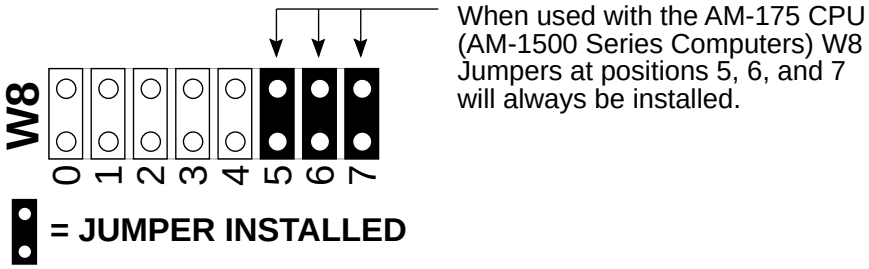
1. **Timing Detection** - The W7 timing detection jumpers must be set based on the type of CPU board used in your computer. Figure 2 shows all supported W7 jumper configurations:



MAC788

Figure 2

2. **Board Address Range**. Memory addresses on the AM-740 can start at any 4MB address boundary, allowing sequential addressing up to 1 gigabyte. Figure 3 shows how the W8 jumpers are used to configure the AM-740 board's starting address. The AM-740 board is configured the same for all CPU board types, with one exception; when used with the AM-175 board, the AM-740 board's W8 jumpers at positions 5, 6, and 7 **will always be installed. For all other CPU boards, W8 jumpers at positions 5, 6, and 7 will always be removed**.



In AM-1500 systems, if total memory in your system exceeds 16MB, only 16MB will be available. Memory residing above the 16MB limit is automatically disabled. Also, before an AM-740 board can be installed in an AM-1500 computer, there must already be 4MB of existing memory made up of or either (or both) AM-706 or AM-730 memory boards.

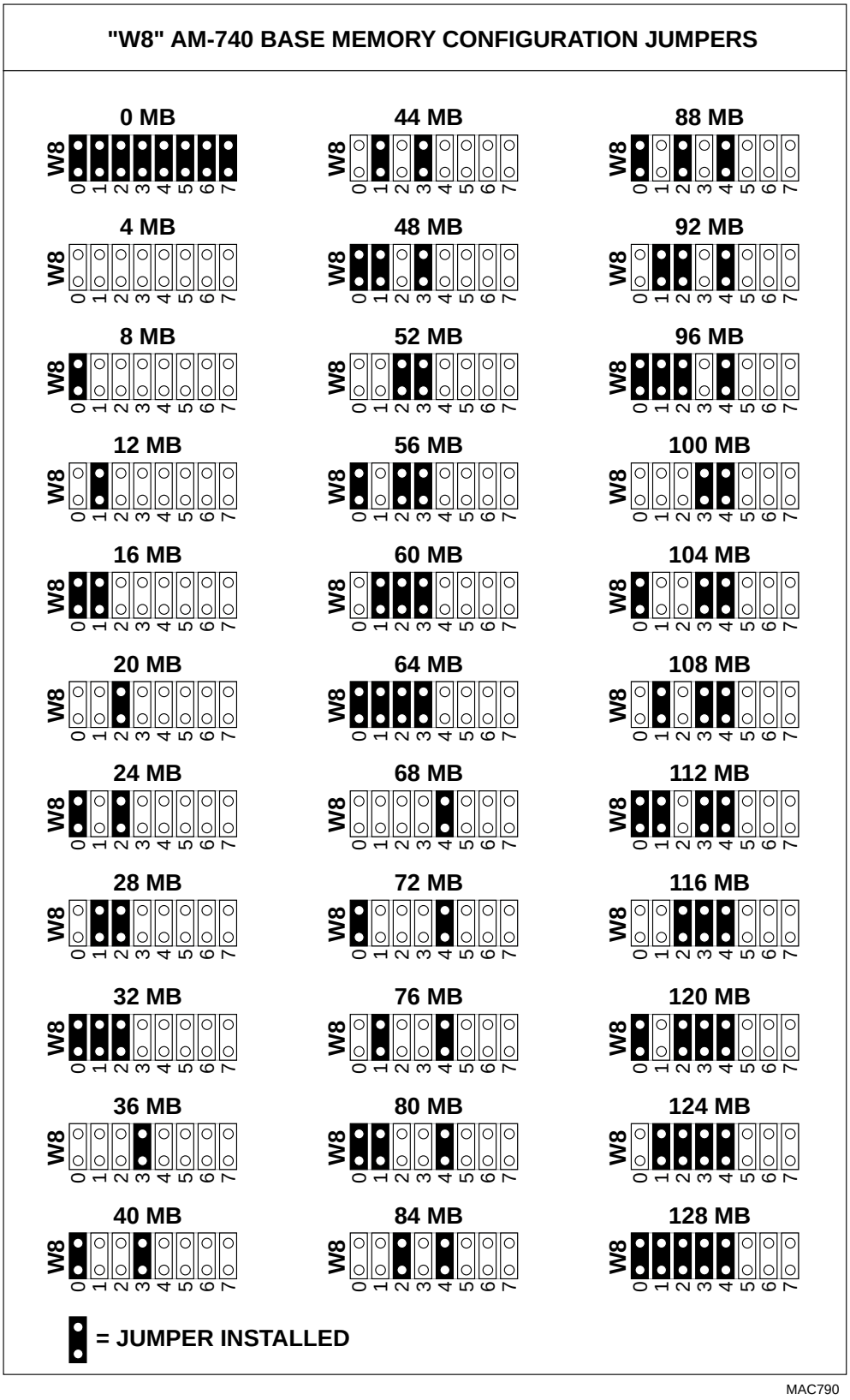


Figure 3. AM-740 Base Address Memory Jumper Configurations

3. **Parity Error** - Parity error interrupt is enabled by setting bit-5 of the I/O register. Bit-5 is set by typing:

   PARITY RETURN

   after the system has booted; but most likely you will want to include the PARITY command as part of your system initialization command file, insuring it is executed each time the computer boots. After the parity error enable bit is set, a parity error will assert the parity error line. The parity status LED, shown in Figure 1, will be lit when a parity error is detected. Once lit, the LED will stay on until the computer is reset.

4.2 Opening the VMEbus Systems

1. Follow the instructions in your VME computer owner's manual that describe how to open the system and gain access to the VME motherboard.

2. Once the computer is open, locate the empty slot that you will be plugging the AM-740 memory board into; or slots if you are going to install more than one AM-740 board.

3. You do not have to remove any IACK or Bus Grant jumpers from the VME motherboard. Simply slide the AM-740 board into position and seat the board into the VME connectors using the nylon insertion/ejection tabs on the AM-740 board.

5.0 ENABLING PARITY ERROR DETECTION FOR THE AM-740

To enable the AM-740 memory board's parity error detection feature, you must use the PARITY command. Although you may use this command at AMOS command level, you will almost always want to use it within your AMOS initialization command file. (For information on changing your AMOSL.INI or AMOS32.INI command file, see the "The System Initialization Command File" in the System Operator's Information section of the ***AMOS/L System Operator's Guide***.)

Place the PARITY command in your AMOSL.INI or AMOS32.INI command file after the first TRMDEF statement.

```
TRMDEF TERM0,AM355=0:9600,ALPHA,100,100,100
PARITY
```

6.0 ACCESSING THE NEW AM-740 MEMORY

Once the AM-740 board (or boards) are installed, the job is complete. Power-up your computer and the AMOS operating system will automatically make the new memory available.

After the new memory has been installed, you will want to verify the computer is detecting all available memory. Enter:

SYSTAT RETURN

Check the resulting display on your terminal screen and insure all the memory has been detected. If your computer does not display the correct amount of memory, double check all the AM-740 configuration jumpers. Don't forget, if your computer is an AM-1500 series, it will not acknowledge memory beyond 16MB.

7.0 THE OCCURRENCE OF A PARITY ERROR

If the job in whose memory partition the parity error occurred is attached to a real terminal (as opposed to a pseudo-terminal), this message appears:

```
?Memory parity error
```

AMOS then halts that job. Also, a "9" appears on the status display panel when a parity error occurs.

When a parity error is reported, the System Operator should inform all users still running that a memory error has occurred, and ask them to finish up or save what they are doing on the system and logoff. Then the System Operator should open the computer and inspect each AM-740 board.

When a parity error occurs, you can identify which memory board encountered the error by looking at each AM-740 memory board to see if its LED indicators are lit. Make note of the memory board on which the parity error occurred.

At this point you must decide whether to continue operating with the memory board that generated the parity error or whether to swap the board out for another one. Parity errors can reflect a transient problem that surfaces once and is never seen again. You may therefore wish to continue running the system as-is until a parity error occurs again. If a parity error occurs on the same memory board more than once or twice in a great while, you probably will want to remove the questionable memory board and return it to your dealer for inspection and/or repair.

To reset an AM-740 memory board's parity error indicator, you must perform a hardware reset on your system. (Note that the REBOOT command, which reboots your system will not reset the AM-740, since it does not perform a hardware reset.)

APPENDIX A

AM-740 PAL KITS

All AM-740 boards are shipped from the factory with a DWP-00740-09 PAL compatible with AM-2000 and AM-3000 computers. If you are installing an AM-740 board into an AM-1500 computer, you must replace the DWP-00740-09 PAL with a DWP-00740-01 PAL. The DWP-00740-01 PAL is included in an upgrade kit, available under the part number PDB-00740-50.

A.1 REPLACING THE PAL FOR AM-1500 APPLICATIONS

Figure A-1 shows the location of the DWP-00740-09 PAL. Carefully remove the PAL by lightly prying up on each end of the chip until you are able to lift it from the socket. Replace the PAL you removed with the DWP-00740-01 PAL included in the upgrade kit.

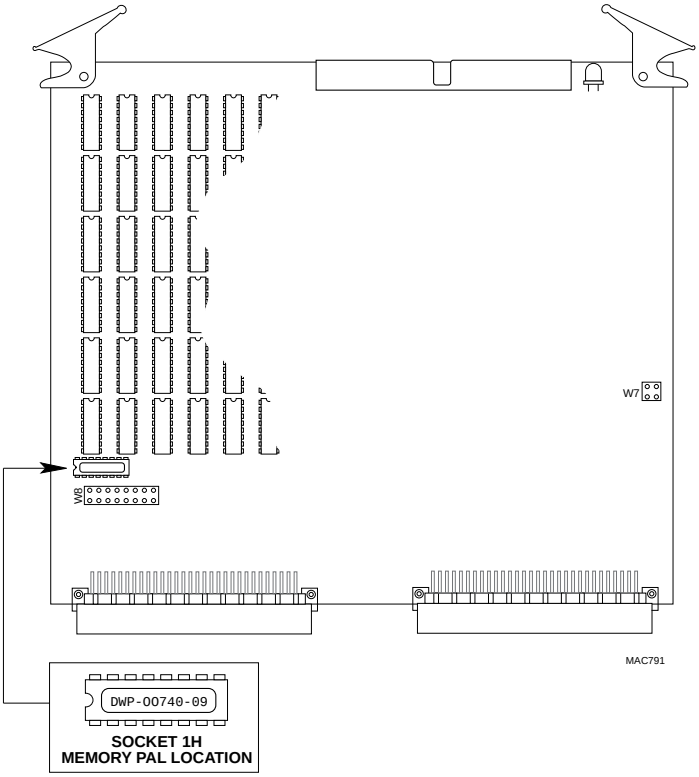


Figure A-1. AM-740 Memory PAL Location